Complex Terrain Meteorology

Federico Giacobbe e Chiara Bindandi

Anno accademico: 2025/2026

1 Characterization of valley flow regimes

In the following section are analysed data from tethered balloons, measured between May 20 and 21, 2013, in Dugway Proving Ground, Utah.
The terrain types present in this place are Playa and desert shrub and for this reason large energy budget contrasts occur between the Playa and its drier surroundings. In general, the Playa is cooler during the day and warmer at night than the surrounding desert-shrub area. These surface contrasts drive "playa breezes", off-playa breeze during the day and on-playa breeze during the night. These breezes correspond to up-valley flows during the day and down-valley flows at nighttime, respectively.

Data files are named according to the UTC time they were measured at; UTC time have been converted to local time to distinguish different parts of the day. By doing so, the days were divided into 4 different phases:

- "Sunset" refers to the hours before sunset, from 23:26 UTC, May 20 to 02:46 UTC, May 21
- "Early night", from 02:55 UTC to 07:08 UTC, May 21
- "Deep night", from 07:21 UTC to 12:27 UTC, May 21
- "Sunrise" refers to the hours after sunrise, from 12:43 UTC to 13:47 UTC, May 21

Looking at the Figure 1, starting from the hours before Sunset, neutral temperature potential profiles are observed, associated with winds coming from the NW[1] quadrant, from Playa to valley, associated with up-valley flows. This direction is completely different from the other regimes which are characterised by winds coming approximately from the S quadrant, from valley to Playa, associated with down-valley flows.
After Sunset, vertical temperature profiles show the formation of a 15-meter-deep inversion layer ("Early night"), which increases as the night progresses ("Deep night") until reaching, before Sunrise, 50 m in height. Below 50 m the inversion layer is intense and above 100 m a neutral profile is observed. As the night progresses, the surface air layer changes its temperature by approximately 10°C, going from 29°C to 19°C.
During Sunrise, considering the red vertical temperature profiles in Figure 1, an increase in potential temperature is observed in the first meters of air over surface. The inversion is broken by surface warming and a neutral potential temperature profile reaches 100 m of height.

Considering the vertical profiles of wind direction and speed, during "Sunset" the wind mainly comes from the N-NW with a relatively constant vertical profile and low intensity, maximum 4 m/s. During the first part of the night, the wind changes direction and blow from S-SW

[1] Convention on abbreviations for cardinal points: N for north, NE for north-east, E for east, SE for south-east, S for south, SW for south-west, W for west and NW for north-west.

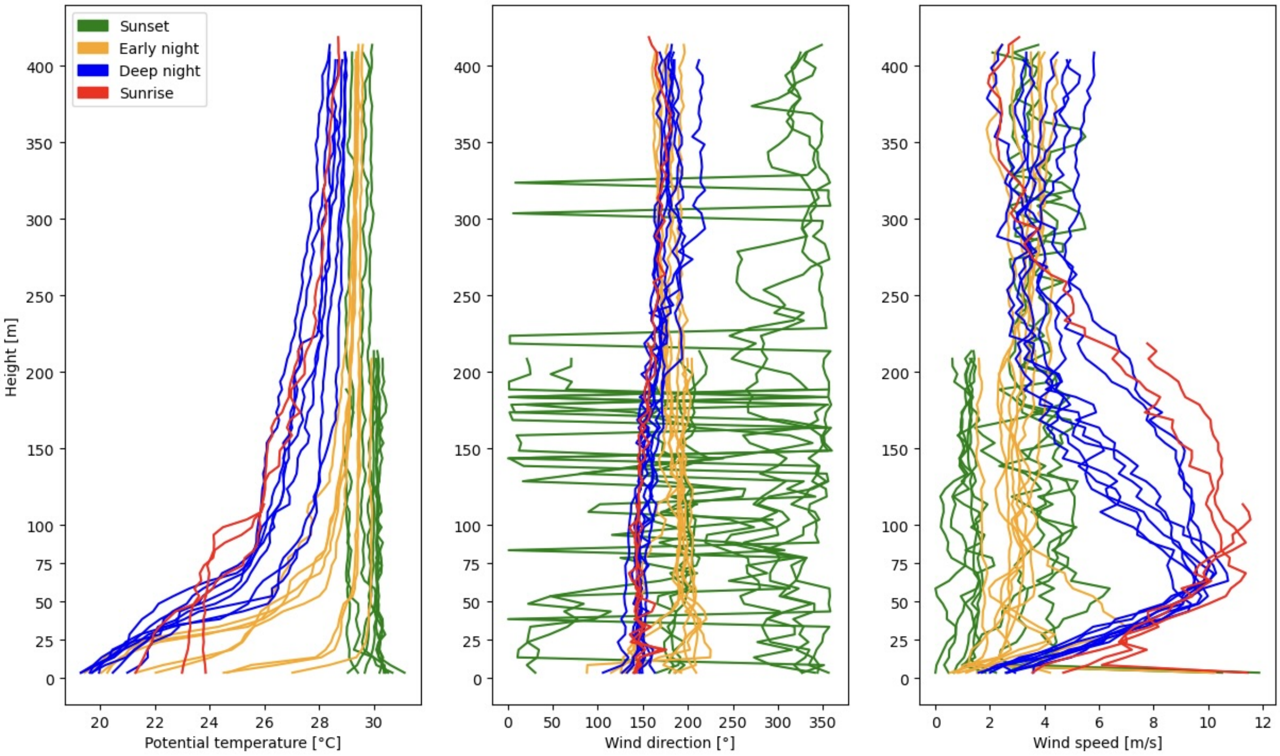

Figure 1: *Vertical profiles of potential temperature, wind direction and speed. Different colours identify different phases of the day.*

with minimal variability in direction and still reduced intensity, the maximum speed reached is around 7 m/s at a height of about 30 m.
As night progresses ("Deep Night"), the direction of origin shifts towards S-SE with limited variability in the direction. In this case, the intensity increases, and the characteristic nose-shaped profile of the downflow also appears, reaching values between 10 m/s and 11 m/s at heights between 60 m and 70 m.
During sunrise, the profile's shape and intensity remain the same as during Deep night.
No further assumption can be made as no data are available for the following hours, but it can be assumed that the vertical profiles return to conditions typical of a convective layer.

1.1 Conclusions

After analysing the data, 4 different time intervals were identified, Sunset, Early night, Deep night and Sunrise.
The first interval is characterized by vertical potential temperature profiles that describe a convective atmosphere and by vertical profiles of wind direction and speed indicating the presence of wind coming from N-NW with low wind speed; these breezes correspond to up-valley flows.
Early night is described by wind coming from the S-SW with weak wind speed and vertical potential temperature profiles denote a decrease of potential temperature and the presence of a forming inversion layer. As the night progresses, wind speed increases and the characteristics nose-shaped profile of the downflows appears; from vertical profiles it is possible to observe a fully formed inversion layer.
The last interval shows the breaking of the inversion layer due to the increase in potential temperature and the subsequent formation of a neutral layer; wind speed and direction remain the same as during Deep night.

Local regimes in absence of synoptic forcing are confirmed. After Sunset, the boundary layer changes from a Convective boundary layer, with neutral potential temperature profile, to a stable one. The inversion starts to grow from the ground and reaches maximum depth just before Sunrise; after Sunrise, the inversion layer breaks quicker than the time it took to form - about one hour to create a 100-meter-deep convective layer.
Night-Day reversal of wind regime is clearly observed: up-valley, coming from NW before Sunset and down-walley, in the opposite direction during night. Despite the nose-shaped profile of the downflows takes time to form, appearing late at night, wind direction changes very quickly and moves to the southern quadrants immediately after Sunset. This confirms the high sensitivity of down-valley winds to local thermal condition.

2 Sonic data pre-processing

In the following Section 5 dataframes containing raw data of wind components (u, v e w) and sonic temperature (Ts) are evaluated, coming from sonic anemometers placed at heights of 0.5 m, 2 m, 5 m, 10 m, 20 m. All the dataset start at 00:30:00 of 02-10-2012 and end at 04:30:00 02-10-2012, so if the sampling rate is 20 Hz, 288001 samples are expected.
The first step is to identify the time gaps, they are shown in Table 1 under "Missing points", they are all around 0.1% for all vertical levels.
In order to have an equal and complete datset for all vertical levels (288001 samples) , time gaps are filled with "Not A Number".
Under "NaN points", in Table 1 , NaN values already present in the raw datasets are counted. Afterwards, data that can be associated with non-physical and anomalous values, that are significantly different from the data distribution, are eliminated. In order to do so, fixed threshold values for the velocity components and sonic temperature are imposed. Values of horizontal wind speed above 30 m/s are set as NaN, as are values for vertical speed above 4 m/s and sonic temperature above 310 K (30°C for 2 m level).
The number of points that exceed the applied thresholds is found in the "Out of range" row in Table 1. It is observed that this value is the same for all the components of the considered data-frame and the percentage of "Out of range" in the datasets is very low and corresponds to 0.002% for 0.5 m, 0.006% for 2 m, 0.004% for 5 m, 0.003% for 10 m and 0.003% for 20 m.
Another method used to remove a single or a small group of data points, characterized by a magnitude too different compared to the data distribution (spikes), is based on the use of moving windows. In this method, called despiking, thresholds are dynamically computed based on the local mean or median and on an acceptable range determined by the variability of the time series around a specific time point.
The type of despiking used in this review is based on the method described by Vickers and Mahrt and assumes that each interval within the dataset follows a Gaussian distribution of independent data characterized by mean and standard deviation (σ).
The moving window is moved along the time series, and at each step, mean and standard deviation are computed. Values of horizontal wind speed and sonic temperature exceeding 3.5σ are marked as possible spikes, as 5σ for vertical wind speed. Each spike can be one or more consecutive measures, and it is replaced by a linear interpolation from its closest neighbours only if possible spike contains less than 4 consecutive points.
The percentage of points changed because marked as spikes is different for each data-frames and for each components of the data-frames, as can be seen in Table 1 under "Changed with despiking in %". The total percentage don't vary much through the sonic anemometers, being $0,15\%$ for 0.5 m, $0,21\%$ for 2m and 5m, $0,14\%$ for 10m and $0,10\%$ for 20m. Overall, in absolute

numbers, the data rejected with despiking is much higher than the out-of-range data, but both represent a minimal percentage of the entire dataset.

	0.5m	**2m**	**5m**	**10m**	**20m**
Missing points	360	300	420	300	280
NaN points	28	23	19	22	18
Out of range	6	16	11	10	10
Changed with despiking	u: 106 v: 196 w: 82 Ts: 61	u: 129 v: 159 w: 53 Ts: 277	u: 68 v: 153 w: 44 Ts: 337	u: 23 v: 148 w: 10 Ts: 221	u: 45 v: 95 w: 16 Ts: 131
Changed with despiking in %	u: 0.04 v: 0.07 w: 0.03 Ts: 0.02	u: 0.04 v: 0.06 w: 0.02 Ts: 0.10	u: 0.02 v: 0.05 w: 0.02 Ts: 0.12	u: 0.008 v: 0.05 w: 0.003 Ts: 0.08	u: 0.02 v: 0.03 w: 0.006 Ts: 0.05

Table 1: *Statistics of processing.*

Looking at the dataset after the all process, some time gaps lasting from 50 ms to 9 s are found. The time gaps are 12 for each vertical level, as shown in Table 2, they do not have the same duration and their start and end times are not the same. This makes it impossible to apply a single interpolation to completely fill the dataset. However, once the phenomenon to be investigated has been chosen, it is possible to interpolate over a sufficiently small time window in order not to remove the variability of the phenomenon and not to introduce artificial data.

	0.5m	**2m**	**5m**	**10m**	**20m**
Total missing points	394	339	450	332	308
Number of time gaps	12	12	12	12	12
Max time gaps	7s	5s	9s	5s	5s
Min time gaps	50ms	50ms	50ms	100ms	50ms

Table 2: *Remaining time gaps*

3 Analysis of sonic data collected at a slope site

In the following section are analysed data, from a measurement campaign conducted in Granite Peak area, in the time interval between 00:00 UTC, 28 September 2012 and 23:55 UTC, 2 October 2012.
These data come from a meteorological tower (ES5) in which are installed 5 sonic anemometers and 5 thermo-hygrometers, placed at height of 0.5 m, 2 m, 5 m, 10 m and 20 m; in addition there is a barometer at 0.5 m.
Sonic data are available as 5-minute averages and distributed across 5 vertical levels; in order

to obtain a single dataset, all the data from the various levels have been merged by date.

Looking at the complete sonic dataset, missing values are founded at 2 m, from 23:30 UTC, 28 September 2012 to 23:55 UTC, 2 October 2012 in all the variable; at 0.5 m 2 records are missing, at 23:30 UTC, 28 September 2012 and at 23:25 UTC, 1 October 2012; there are no time gaps at other levels.

3.1 Downslope flows

Considering local geography and the physic of the phenomenon, where Granite Peak is located to the W, a down-slope at night in the W quadrant would be expected.
To find down-slope flow, true wind direction over time are plotted in Figure 2 and sunset and sunrise time are highlighted (dashed lines), in Table 3, for every day.

Day	Sunrise UTC	Sunset UTC
2012-09-28	2012-09-28 13:27	2012-09-28 01:20
2012-09-29	2012-09-29 13:28	2012-09-29 01:19
2012-09-30	2012-09-30 13:29	2012-09-30 01:17
2012-10-01	2012-10-01 13:30	2012-10-01 01:15
2012-10-02	2012-10-02 13:31	2012-10-02 01:14

Table 3: *Sunset and sunrise time in UTC at ES5 location.*

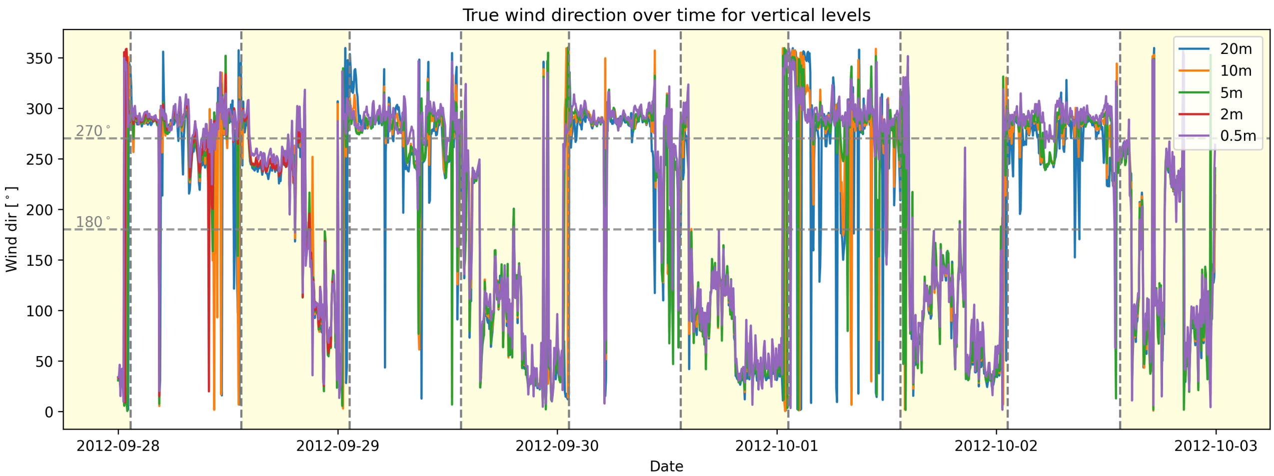

Figure 2: *Time series of wind direction for each vertical level. Yellow background is daytime and white background is nighttime.*

From wind direction time series, the night-day reversal is relevant; indeed, at sunset there is a rapid change in wind direction. Probably the actual sunset in ES5 is slightly earlier, due to position of Granite Peak, in fact the switch in wind direction occurs slightly to the left of the sunset line in the graph in Figure 2.
During nights westerly winds are dominant, alternating with some spikes, probably due to down-valley winds and local interactions. In general, it can be said that the necessary condition to have down-slope wind is present in all available nights. During the day, the up-slope regime is less pronounced but still present, with an easterly direction prevailing in the middle of the day.

3.2 Vertical profiles

To confirm the presence of down-slope wind the along slope wind speed vertical profiles are plotted in Figure 3. Data from each night have been extrapolated from the dataset and all wind speed values collected into one box plot for each level. All available nights seem to have a down-slope flow with the characteristic nose profile; based on the available data, the maximum wind speed is below 3 m/s during the entire period.
Defined z_{jet} as the height at which maximum wind speed is reached, z_{jet} can be identified as equal to or approximately 5 m.

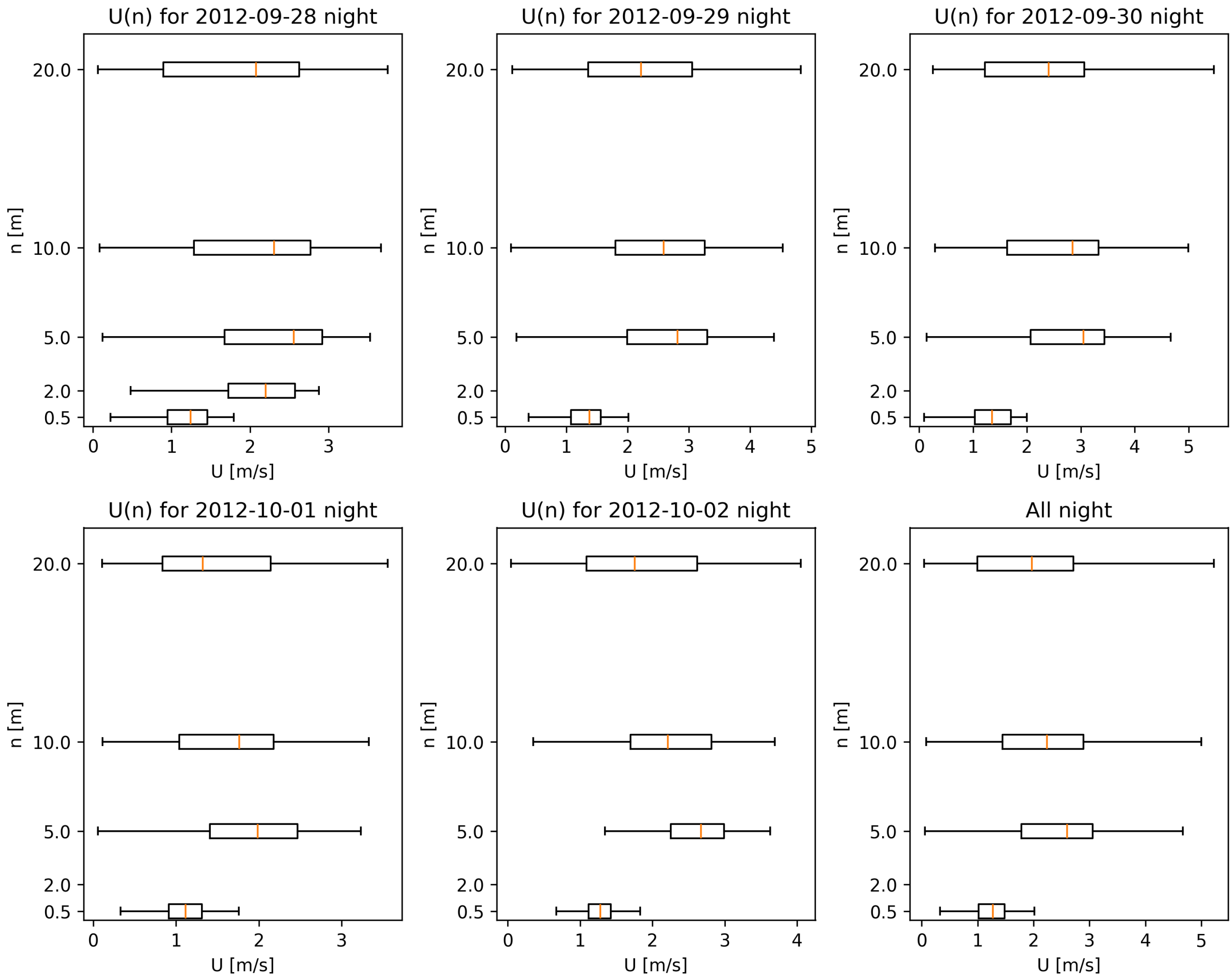

Figure 3: *Nighttime vertical profile of wind speed along the slope. The first 5 plots contain measurements from sunset to sunrise divided by day, the last contains measurements from all available nights.*

To evaluate the stability of the 20-meter-deep layer, potential temperature is taken into consideration, using the equation $\theta = T(1000/p)^{0.286}$. T is the temperature measured by slow sensors at the same levels as sonic anemometers and $p(n)$ is obtained with the hypsometric equation knowing $p(0.5m)$, considering this is a good approximation for 20 m height.
The general trend over several days of vertical profile of potential temperature confirms expectations. During the night in Figure 4, θ increases with altitude, indicating a stable layer near the ground created by cooling. During daylight hours in Figure 5, the potential temperature has a wider range of values, but overall the median outlines a θ profile that decreases slightly with height, indicating a convective or unstable layer.

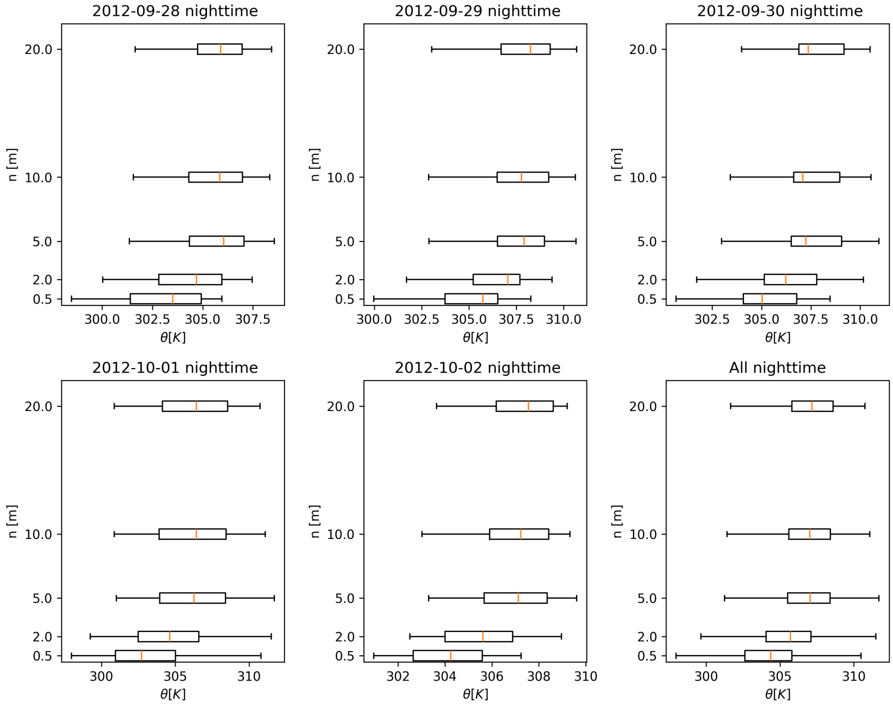

Figure 4: *Nighttime vertical profile of potential temperature. The first 5 plots contain measurements from sunset to sunrise divided by day, the last contains measurements from all available nights.*

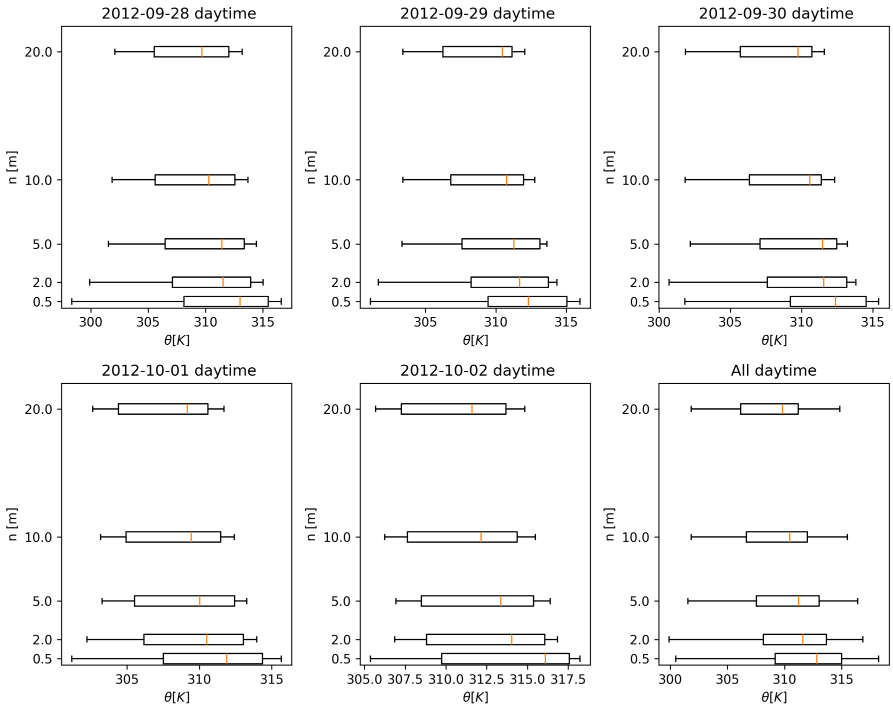

Figure 5: *Daytime vertical profile of potential temperature. The first 5 plots contain measurements from sunrise to sunset divided by day, the last contains measurements from all available daytime.*

3.3 Turbulent variables

Taking theory into consideration, a net downward transport of momentum, $\overline{u'w'}$, must be present below z_{jet}, while above a net upward transport of momentum is present. Looking at Figure 6 this is confirmed.

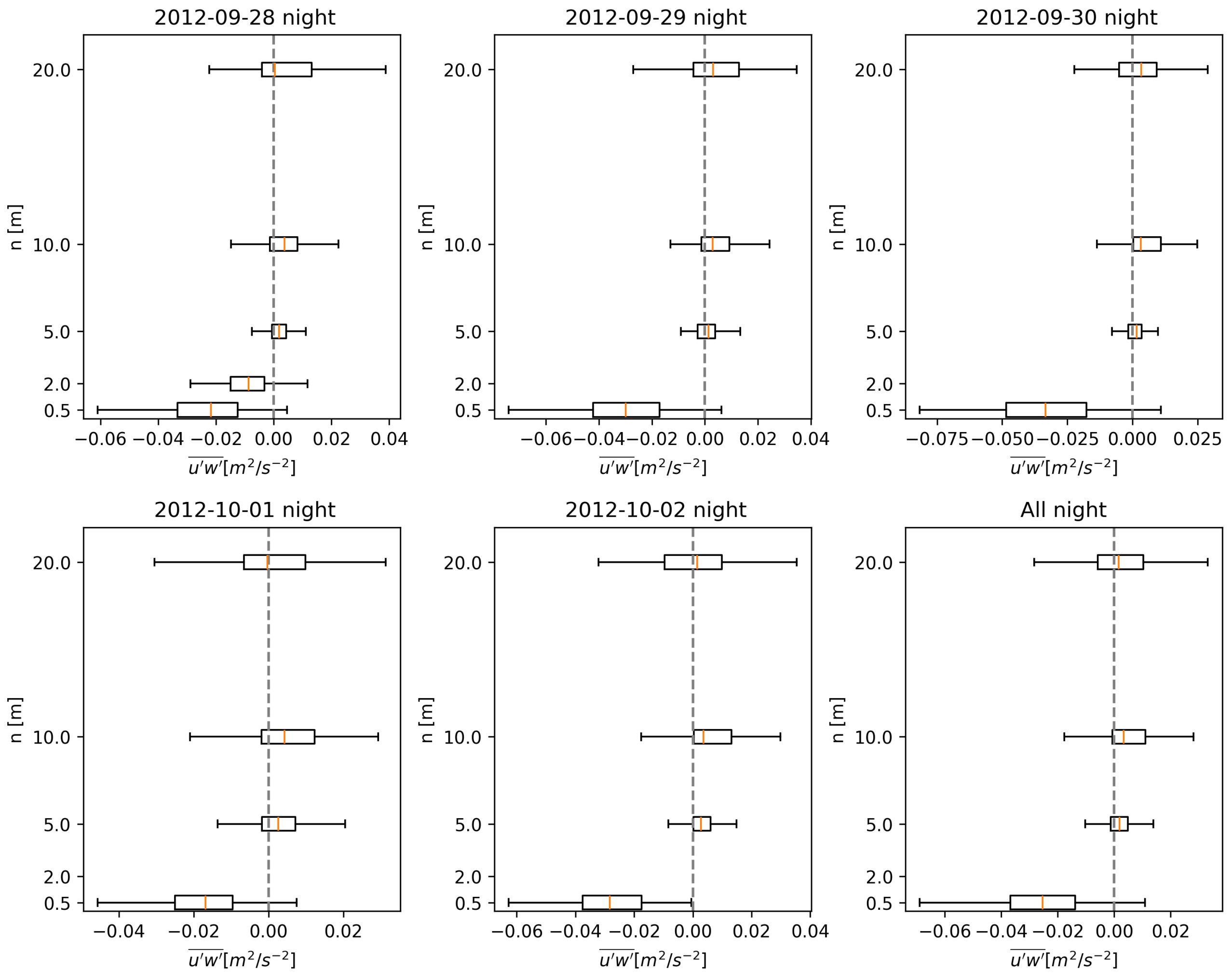

Figure 6: *Nighttime vertical profile of kinematic momentum flux $\overline{u'w'}$. The first 5 plots contain measurements from sunset to sunrise divided by day, the last contains measurements from all available nights.*

It is observed that $\overline{u'w'}$ at 5 m height is positive and it is less than $\overline{u'w'}$ at 10 m, this could indicate that z_{jet} is located slightly below 5 m because $\overline{u'w'}$ is about 0 at z_{jet} .

From the theory, it is known that the sign of sensible heat flux normal to the slope, $\overline{w'T'_s}$, does not vary with height and always remains negative while the sign of the flux along the slope, $\overline{u'T'_s}$, is positive below z_{jet} and negative above; inspecting Figures 7 and 8 these trends are validated.
Taking into account the 5 m height data, it is observed that $\overline{u'T'_s}$ is negative, this could indicate that z_{jet} is located slightly below this height.

The use of sonic temperature instead of potential temperature in sensible heat fluxes is possible because the fluctuations from the mean of the two temperatures are comparable. Moreover the sonic temperature is the only thermodynamic variable available with sample rate of 20 Hz.

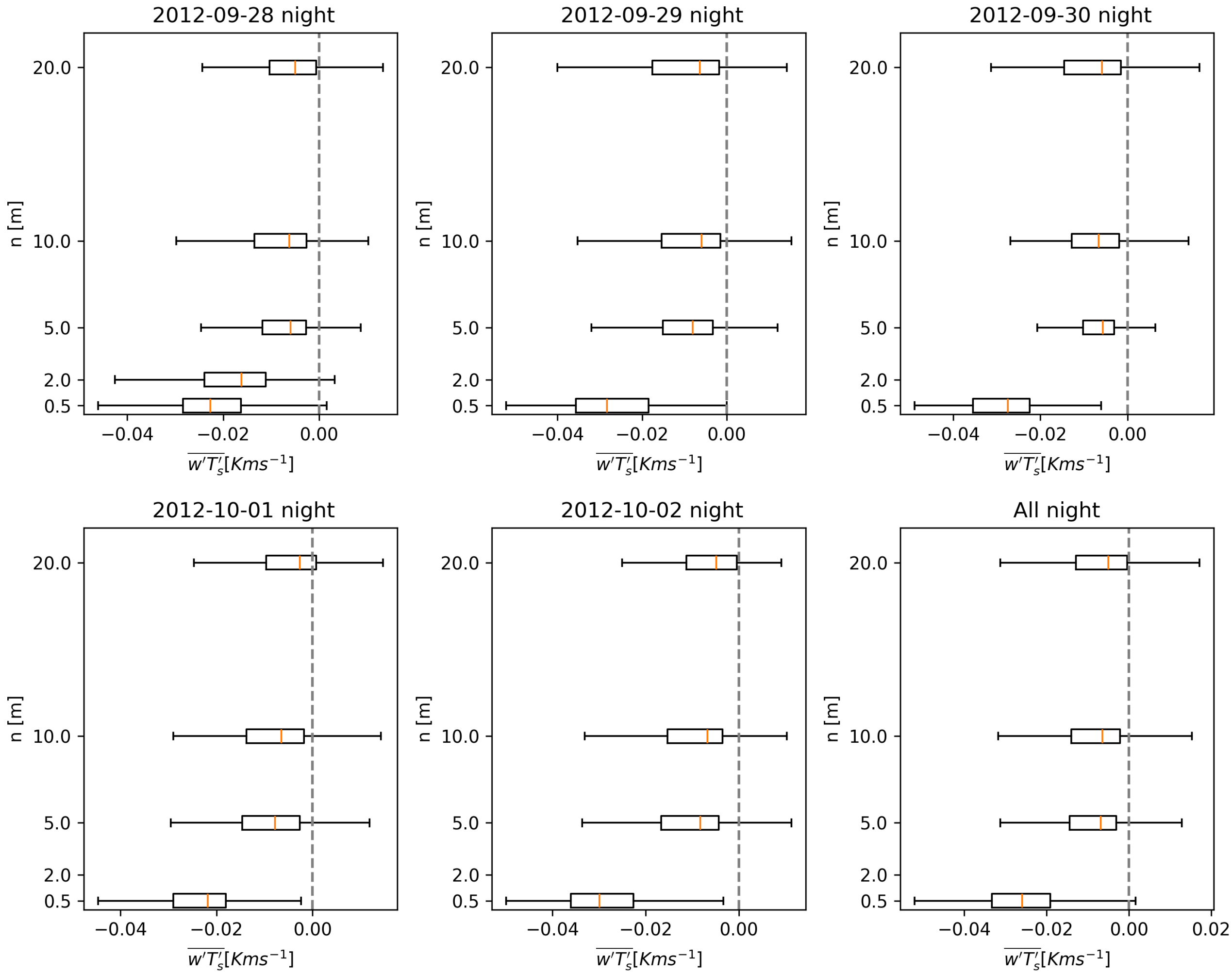

Figure 7: *Nighttime vertical profile of kinematic sensible heat flux* $\overline{w'T'_s}$*. The first 5 plots contain measurements from sunset to sunrise divided by day, the last contains measurements from all available nights.*

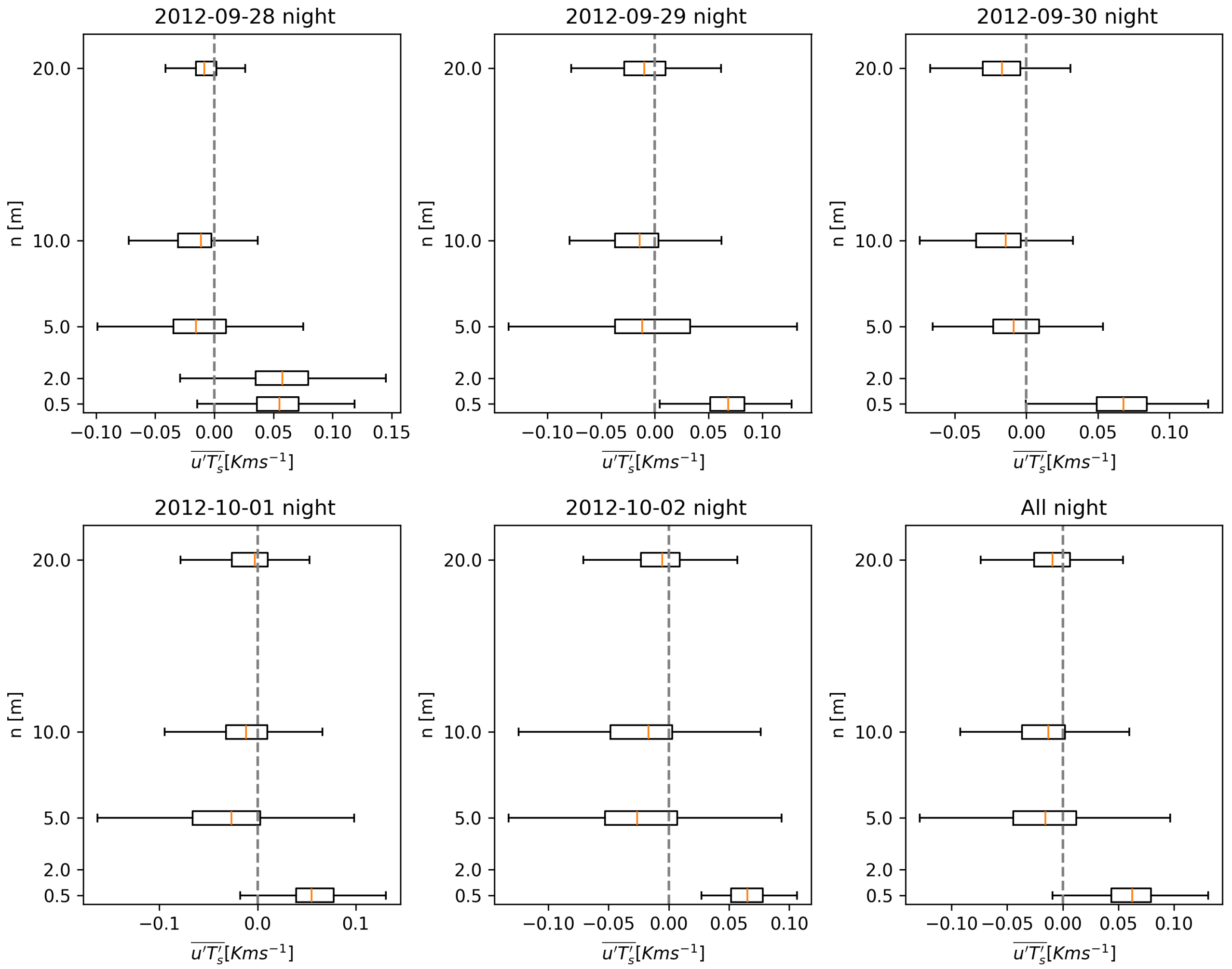

Figure 8: *Nighttime vertical profile of longitudinal kinematic heat flux $\overline{u'T'_s}$. The first 5 plots contain measurements from sunset to sunrise divided by day, the last contains measurements from all available nights.*

Another fundamental quantity to evaluate in this topic is the Total Kinetic Energy, expressed as

$$TKE = 0.5(\overline{u'^2} + \overline{v'^2} + \overline{w'^2})$$

it is plotted in Figure 10 and shows its minimum at 5 m, this is coherent with the typical TKE profile in presence of a low level jet. The jet creates a low turbulent zone near its maximum, unlike what would happen on flat terrain, where TKE decreases with altitude.
Known the TKE budget equation in s-n coordinate system, the buoyancy term

$$P_B = \frac{g}{\theta_0}(\overline{w'T'_s}cos(\alpha) - \overline{u'T'_s}sin(\alpha)) = \frac{g}{\theta_0}B$$

is evaluated with a slope angle $\alpha = 3.6°$. This quantity, in Figure 9, is negative and decreases with n, due to the stable condition established during the night.

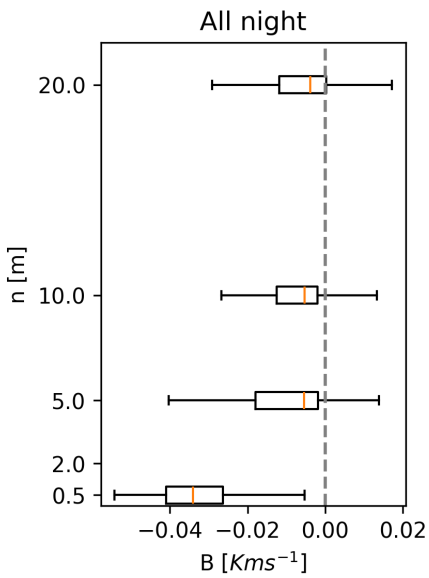

Figure 9: *Vertical profile of B.*

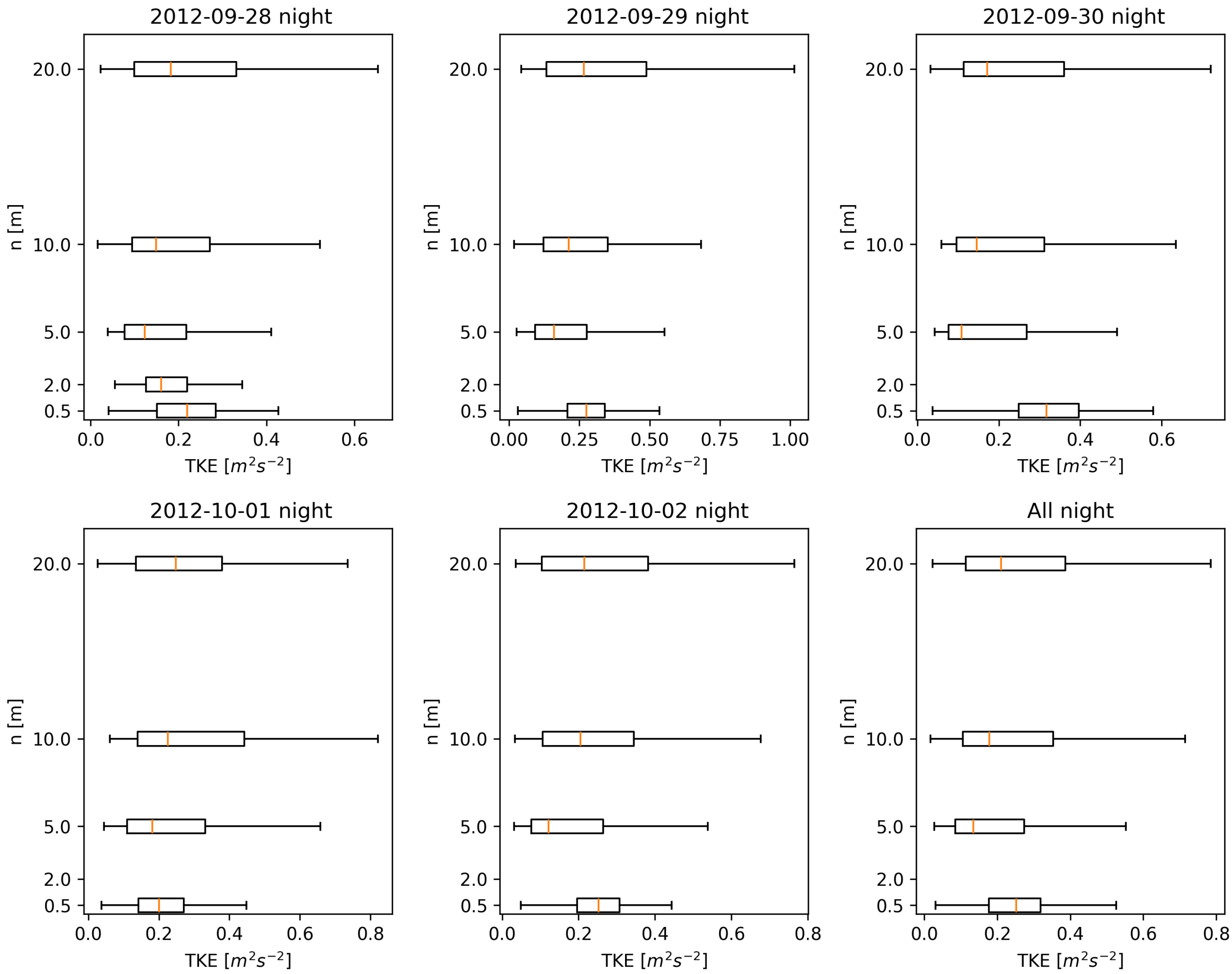

Figure 10: *Nighttime vertical profile of turbulent kinetic energy. The first 5 plots contain measurements from sunset to sunrise divided by day, the last contains measurements from all available nights.*

3.4 Conclusions

In conclusion, it can be said that the wind present at night, which has a maximum speed of about 3 m/s at 5 m height for the entire period analysed, descends along the Granite Peak and furthermore the presence of the nose structure of the wind speed confirms the presence of downslope flows. To underline their presence, turbulent variables, $\overline{u'w'}$, $\overline{w'T_s'}$ and $\overline{u'T_s'}$ are also analysed; it is detected that the data consistently follow what is reported by theory.
In addition to this, TKE was also examinated to see if its minimum was present at the height where the maximum speed of U is found.

Improvements in the experimental campaign may include:

- Increasing the vertical measurement levels, because even though 4/5 levels are sufficient to draw qualitative conclusions, are not sufficient to perform a fit on the nose profile with a theoretical model.

- Using tethered balloons to obtain vertical profiles creates a better classification of the phenomena above the surface. For example, they can identify valley flow regimes, which differ between day and night and are linked to slope flows.

- Having a high-frequency humidity measurement would allow for a more accurate assessment of the potential temperature. The reliability of sonic temperature depends greatly on the relative humidity of the air.